

5.2 Circular motion

There are many examples of objects travelling at constant speed in circles, e.g. planets, artificial satellites, charged particles in a magnetic field, etc. The physics in all these cases can be described and analysed using the ideas developed by Newton. The concepts in this section have applications in many contexts present in other sections of this specification,

such as planetary motion in section 5.4.3 (HSW1, 2, 5, 9).

This section provides knowledge and understanding of circular motion and important concepts such as centripetal force and acceleration.

5.2.1 Kinematics of circular motion

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) the radian as a measure of angle	M4.7
(b) period and frequency of an object in circular motion	
(c) angular velocity ω , $\omega = \frac{2\pi}{T}$ or $\omega = 2\pi f$	